

CSC 3114 Cryptology and Coding Theory

Revision Question Bank 1

Conventions: Use the English alphabet A–Z. For letter ciphers, ignore spaces and punctuation unless otherwise stated. Caesar encryption shifts letters forward, with wraparound. For Vigenère calculations, use $A = 0, B = 1, \dots, Z = 25$ and addition modulo 26. One byte is 8 bits; 1 KiB is 1024 bytes.

A. Cryptography and Classical Ciphers

Question 1: Foundations and security goals [10 marks]

- (a) Distinguish cryptography, cryptanalysis and cryptology. [3 marks]
- (b) Define plaintext, ciphertext and a cryptographic key. [3 marks]
- (c) Explain confidentiality and integrity. Give one example of a violation of each. [4 marks]

Question 2: Encryption and adversaries [10 marks]

- (a) Draw a labelled encryption–decryption model for Alice sending a message to Bob over an insecure channel. Include the key, plaintext, ciphertext and Eve. [4 marks]
- (b) Distinguish symmetric encryption from asymmetric encryption in terms of the keys used and which keys must remain secret. [3 marks]
- (c) Eve modifies Alice’s message and sends it to Bob while pretending to be Alice. Identify two security goals threatened, and explain why keeping the encryption algorithm secret is not the central security assumption in the slides. [3 marks]

Question 3: Caesar and ROT13 ciphers [10 marks]

- (a) Encrypt ATTACK AT ONCE using a Caesar shift of 4. [3 marks]
- (b) Decrypt VHFUHW PHVVDJH using a Caesar shift of 3. [3 marks]
- (c) Explain why applying ROT13 twice recovers the original message. [2 marks]
- (d) Explain why a Caesar cipher is vulnerable to brute-force cryptanalysis, stating the number of non-identity shifts. [2 marks]

Question 4: Substitution techniques [10 marks]

- (a) Distinguish substitution from transposition. [2 marks]
- (b) Compare monoalphabetic and polyalphabetic substitution, using Caesar and Vigenère as examples. [4 marks]
- (c) Explain homophonic substitution and polygram substitution. Give a simple illustrative mapping for each. [4 marks]

A. Cryptography and Classical Ciphers (continued)

Question 5: Transposition ciphers

[10 marks]

- (a) Encrypt HELLO WORLD using the two-row rail fence method: place successive letters alternately in the upper and lower rows, then read the upper row followed by the lower row. Show the arrangement. [3 marks]
- (b) Encrypt DEFEND THE EAST WALL OF THE CASTLE using regular columnar transposition with the keyword GERMAN. Write the letters row by row, pad with X to complete the rectangle, and read columns from top to bottom in alphabetical order of the keyword letters. Show the rectangle and the ciphertext. [5 marks]
- (c) Distinguish regular from irregular columnar transposition. [2 marks]

Question 6: The Vigenere cipher

[10 marks]

- (a) Explain how the keyword is aligned with a message longer than the keyword. [2 marks]
- (b) Encrypt COMPUTING GIVES INSIGHT with the keyword LUCKY. Show the repeated key and letter-by-letter working. [5 marks]
- (c) Use the same key to decrypt the first five ciphertext letters obtained in part (b), showing your working. [3 marks]

Question 7: XOR and the Vernam cipher

[10 marks]

- (a) Write the truth table for XOR and calculate $0101 \oplus 0011$ and $0010 \oplus 1010$. [3 marks]
- (b) Encrypt DOG with the key CAT using bitwise XOR. Use the following 8-bit character codes and give the ciphertext as three binary bytes. [4 marks]

Character	D	O	G	Character	C	A	T
Code	01000100	01001111	01000111	Code	01000011	01000001	01010100

- (c) Recover the first plaintext byte by applying XOR again with the corresponding key byte. Explain why Vernam ciphertext is often displayed as binary, decimal or hexadecimal values. [3 marks]

Question 8: One-time pads and cipher operation

[10 marks]

- (a) State four requirements concerning key length, randomness, reuse and secrecy for a secure one-time pad. [4 marks]
- (b) Explain how a one-time pad differs from a Vigenere cipher that repeats a short keyword, and why randomness matters. [3 marks]
- (c) Distinguish a bit-stream cipher from a block cipher in terms of how the plaintext is processed. Name two operations that may be used in a block cipher. [3 marks]

B. Introduction to Coding Theory

Question 9: Purpose and applications of coding theory [10 marks]

- (a) Define coding theory and distinguish its main purpose from that of cryptography. [3 marks]
- (b) Explain why errors can arise both when information is transmitted and when it is stored for later retrieval. [3 marks]
- (c) Identify two applications of error-correcting codes from the slides and explain the benefit in each case. [4 marks]

Question 10: The communication system [10 marks]

- (a) Draw a labelled digital communication model showing the source, source encoder, channel encoder, modulator, noisy channel, demodulator, channel decoder, source decoder and sink, in the correct order. [5 marks]
- (b) Distinguish source coding from channel coding. [3 marks]
- (c) Explain the roles of the modulator and demodulator. [2 marks]

Question 11: Error detection and error correction [10 marks]

- (a) Explain how redundancy can help a receiver detect or correct transmission errors. [3 marks]
- (b) Distinguish backward error correction (BEC) from forward error correction (FEC), including what the receiver does after an error is detected. [4 marks]
- (c) Explain the basic idea of a parity bit and why detecting an error does not necessarily identify which bit is wrong. [3 marks]

Question 12: A three-bit error-detecting code [10 marks]

The source codes are apple $\rightarrow$ 00, banana $\rightarrow$ 01, cherry $\rightarrow$ 10, grape $\rightarrow$ 11. The channel encoder uses

$$00 \rightarrow 000, \quad 01 \rightarrow 011, \quad 10 \rightarrow 101, \quad 11 \rightarrow 110.$$

- (a) Explain why receiving 01 after transmitting the source word 00, without channel encoding, need not reveal that an error occurred. [2 marks]
- (b) List all words obtainable from 000 by exactly one bit error. Explain how the channel code detects each error. [3 marks]
- (c) If 100 is received and exactly one bit error occurred, list every possible transmitted codeword. Explain why correction is ambiguous. [3 marks]
- (d) Calculate the code rate and state the number of redundant bits per encoded message. [2 marks]

B. Introduction to Coding Theory (continued)**Question 13: A five-bit error-correcting code****[10 marks]**

Consider the channel encoding scheme

$$00 \rightarrow 00000, \quad 01 \rightarrow 01111, \quad 10 \rightarrow 10110, \quad 11 \rightarrow 11001.$$

- (a) List all five received words resulting from exactly one bit error in 00000. **[2 marks]**
- (b) For the received word 10000, count the differing bit positions relative to each valid codeword. Assuming at most one error, identify the transmitted codeword and recover the two-bit source word. **[4 marks]**
- (c) Determine the minimum distance by comparing the valid codewords. Show the pairwise counts. **[3 marks]**
- (d) Calculate the code rate. **[1 marks]**

Question 14: Repetition coding and majority decoding**[10 marks]**

A two-bit source word is repeated five times, as in $01 \rightarrow 0101010101$.

- (a) Encode the source word 10. **[2 marks]**
- (b) Decode 1100100010 by applying majority voting separately to odd-numbered and even-numbered positions. Show both groups and the recovered word. **[4 marks]**
- (c) State how many bit errors this scheme is guaranteed to correct and justify the answer using majority voting. **[2 marks]**
- (d) Calculate its code rate and explain one disadvantage of repetition coding. **[2 marks]**

Question 15: Redundancy and transmission efficiency**[10 marks]**

- (a) A source word contains k bits and is repeated $2r + 1$ times. Express the transmitted length and code rate in terms of k and r . **[3 marks]**
- (b) For $k = 2$ and $r = 3$, determine the transmitted length, number of redundant bits and guaranteed number of correctable bit errors. **[3 marks]**
- (c) Compare the rates of the two-to-three-bit code, the two-to-five-bit code and fivefold repetition of a two-bit word. Explain the trade-off illustrated by these examples. **[4 marks]**

Question 16: Channel coding concepts and design goals**[10 marks]**

- (a) Define minimum distance and error-correction capability. **[2 marks]**
- (b) Explain channel capacity and its significance for communication over a noisy channel. **[2 marks]**
- (c) State four goals to consider when designing channel encoders and decoders. **[4 marks]**
- (d) Describe two practical benefits of improved error-correcting codes discussed in the slides. **[2 marks]**

C. Steganography

Question 17: Steganography, cryptography and watermarking [10 marks]

- (a) Define steganography and compare its primary goal and visible output with those of cryptography. [3 marks]
- (b) Distinguish steganography from digital watermarking in terms of purpose and the relationship between the hidden information and its carrier. Give one example of each. [4 marks]
- (c) Explain the separate roles of encryption, embedding and authentication when used together to protect a hidden message. [3 marks]

Question 18: Embedding and extraction [10 marks]

- (a) Define cover object, payload, stego-key and stego-object. [4 marks]
- (b) Draw a labelled embedding and extraction model. Write equations for producing the stego-object and recovering the payload, and state the condition for correct recovery. [4 marks]
- (c) Explain why relying only on secrecy of the embedding program is weaker than using a secret stego-key. [2 marks]

Question 19: Historical and text-based techniques [10 marks]

- (a) Describe two historical steganographic methods presented in the slides. Explain how each conceals information. [4 marks]
- (b) Recover the hidden message by taking the first letter of each word in the following sentence: [2 marks]

Since Everyone Can Read, Encoding Text In Neutral Sentences Is Doubtfully Effective.

- (c) Suppose one space encodes 0 and two spaces encode 1. Represent 10110 as a sequence of five gap widths. State three additional details the receiver needs for unambiguous extraction. [4 marks]

Question 20: Least-significant-bit image embedding [10 marks]

An RGB pixel has component values (201, 118, 45), represented by

$$R = 11001001, \quad G = 01110110, \quad B = 00101101.$$

- (a) Explain the principle of least-significant-bit (LSB) embedding and why small changes to image components may be difficult to notice. [3 marks]
- (b) Embed the payload bits 010 by replacing the LSB of R, G and B respectively. Show the resulting binary values and decimal RGB triple. [4 marks]
- (c) Extract the three payload bits from your modified pixel. Explain why a visually ordinary image is not necessarily free of hidden data. [3 marks]

C. Steganography (continued)

Question 21: Image capacity and file formats [10 marks]

- (a) Write the raw image-payload capacity formula in terms of width, height, number of colour channels and embedded bits per channel. Define the units of the result. [2 marks]
- (b) Calculate the raw capacity of a 1024×768 RGB image when one LSB per channel is used. Give your answer in bits, bytes and KiB. [3 marks]
- (c) State two reasons why usable payload capacity may be smaller than raw capacity. [2 marks]
- (d) Explain why the image format matters to payload survival. Compare PNG pixel preservation with JPEG quantization, and identify a risk of embedding through GIF palette indices. [3 marks]

Question 22: Audio and other embedding approaches [10 marks]

- (a) Briefly explain LSB coding, echo hiding, phase coding and spread spectrum in audio steganography. [4 marks]
- (b) Explain how resampling, noise reduction and lossy compression can affect hidden audio payloads. [3 marks]
- (c) Explain the purpose of a mimic function in steganography and how changing a message's statistical profile may help it avoid automated attention. [3 marks]

Question 23: Carriers, applications and channel risks [10 marks]

- (a) Identify two places where a payload may be embedded in video and two features that can carry hidden information in network communication. [4 marks]
- (b) Explain how video transcoding and network normalization may disrupt these hidden channels. [2 marks]
- (c) Give two legitimate applications and two potential misuses of steganography identified in the slides. [4 marks]

Question 24: Steganalysis and detection errors [10 marks]

- (a) Define steganalysis. [2 marks]
- (b) Describe structural, visual or auditory, statistical, and machine-learning approaches to steganalysis. [4 marks]
- (c) A detector flags a clean image as containing hidden data and fails to flag another image that does contain hidden data. Name the two types of error and explain each. [2 marks]
- (d) Explain why failure to notice visible or audible changes alone is insufficient evidence that no information has been hidden. [2 marks]